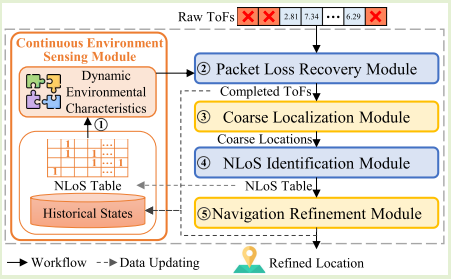
ToF-Based Indoor Localization for Harsh Environments: A Continuous Environment Sensing Framework

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***Abstract*—Nonline-of-sight (NLoS) propagation and packet loss are critical factors that influence the accuracy and robustness of time-of-flight (ToF)-based indoor localization and navigation systems in harsh environments. In this article, we propose a continuous environment sensing framework designed for packet loss recovery and NLoS identification in ToF-based localization systems. The proposed framework comprises a *continuous environment sensing module*, a *packet loss recovery module*, a *coarse localization module*, an *NLoS identification module*, and a *navigation refinement module*. Specifically, the continuous environment sensing module captures dynamic environmental characteristics by updating an NLoS table (from the NLoS identification module) and tracking the target's states (location and velocity). Leveraging the acquired characteristics, we develop a robust low-rank matrix recovery algorithm for packet loss recovery. Subsequently, through coarse localization, we generate localization estimates for different ToF subsets. Based on these localization estimates, we then design a residual analysis-based algorithm for NLoS identification, incorporating an environmentally adaptive residual threshold strategy. Finally, a navigation refinement module is employed to mitigate NLoS errors and refine the trajectory. Experimental results demonstrate that, under a 30% packet loss rate in commodity UWB systems, the proposed framework achieves average localization and navigation accuracies of 17.71cm and 47.42cm, respectively, while reducing the localization error by 51.03% in static scenarios and 50.18% in mobile scenarios compared with state-of-the-art algorithms.**



***Index Terms*—Continuous environment sensing, indoor localization and navigation, nonline-of-sight (NLoS) identification, packet loss recovery, time of flight (ToF).**

NOMENCLATURE

a_n, $\mathbf{x}^n$	ID/Location of the nth anchor.
$\mathbf{x}_k$, $\tilde{\mathbf{x}}_{k,s}$, $\hat{\mathbf{x}}_k$	Target's real/coarse estimated/refined estimated location.
$\mathbf{r}_k$, $\tilde{\mathbf{r}}_k$, $\hat{\mathbf{r}}_k$	Raw/Completed/Estimated ToFs.
$\mathbf{A}_k$, $\hat{\mathbf{A}}_k$	Raw/Estimated ToF matrix.
$\mathbf{A}_k^{\text{comp}}$,$\mathbf{A}_k^{\text{los}}$	Completed/LoS ToF matrix.
$\mathbf{M}_\Omega$, $\mathbf{M}_{\Omega_{\text{nlos}}}$	Observation mask matrix/NLoS table.
N, K, S	# All anchors/Time window/Localization subsets.
n, k, s	Anchor's/Time/Subset index.
ξ, h	Threshold coefficient/Smoothing parameter.

Received 15 August 2025; revised 8 September 2025; accepted 11 September 2025. Date of publication 22 September 2025; date of current version 16 October 2025. This work was supported by the National Natural Science Foundation of China under Grant 62171086. The associate editor coordinating the review of this article and approving it for publication was Dr. Yun-Ju Lee. *(Corresponding author: Xiansheng Guo.)*

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Digital Object Identifier 10.1109/JSEN.2025.3610128

I. INTRODUCTION

INDOOR location-based Services (ILSs) have a wide range of applications, including asset tracking [1], monitoring services [2], and automated valet parking [3], [4], [5]. These services rely on various indoor localization technologies, such as time of flight (ToF) [6], time difference of arrival (TDoA) [7], angle of arrival (AoA) [8], and received signal strength indicator (RSSI) [9], [10], [11]. However, RSSI is highly susceptible to signal attenuation and multipath effects, while AoA and TDoA require specialized hardware or precise synchronization. In contrast, ToF directly measures signal propagation time, providing a simple yet highly accurate solution under

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ideal conditions [12]. However, in harsh environments, its performance can still be affected by multipath effects, signal attenuation, and interference.

In harsh environments, ToF-based localization suffers from two main challenges: non-line-of-sight (NLoS) propagation and **packet loss**. The former occurs when signals are blocked or reflected by obstacles, resulting in increased ranging errors between devices and ultimately leading to localization inaccuracies [13], [14]. The latter mainly results from signal fading and signal collisions. Signal fading occurs due to long propagation distances or large obstacles, leading to severe attenuation and eventual packet loss [15]. Signal collisions occur in high-concurrence transmission, where multiple localization targets transmit at the same time to receivers sharing the same time–frequency resources, leading to interference and data loss [16].

NLoS identification is key to minimizing the impact of NLoS propagation on ToF-based localization systems [13], [14]. Methods with the highest detection accuracy rely on the channel impulse response (CIR) of the signal [17], [18], but these methods often require extensive preparation, such as dataset construction and model training. Some unsupervised methods for NLoS identification have been proposed [19], [20], [21]. For instance, Huang et al. [19] employed a weak labeling method, utilizing map and inertial sensor information to handle unlabeled data. Silva and Hancke [20] employed distance residual instead of channel statistics, a key approach for engineering applications. Building upon [20], Flueratoru et al. [21] conducted residual analysis based on kernel density estimation (KDE) for the unsupervised labeling of CIR data. However, these methods overlooked the influence of the dynamic change of the environment on the NLoS identification. Several NLoS identification methods for dynamic environments have been proposed [13], [22]. Zhu et al. [13] proposed a robust method to identify LoS/NLoS in dynamic environments, but did not address localization. Zhang et al. [23] proposed an evolutionary algorithm based on genetic theory to enhance particle filtering, utilizing 5G ToF and GNSS velocity data, with its performance gains mainly deriving from multisource observations. Su et al. [24] proposed an adaptive scheme based on the channel status identification and ranging residuals to dynamically update the ranging weights. However, the above studies overlooked performance degradation from packet loss, which can alter the dimensions of the ranging frames and render some models inapplicable, potentially leading to localization failure in severe cases.

Packet loss recovery methods can be categorized into two types: neural network-based models [25], [26] and matrix recovery methods [27]. Neural network-based methods train models on datasets to learn spatial correlations for packet loss recovery. Wang et al. [25] proposed a graph-based location mapping network that leverages the spatial homogeneity hypothesis for packet loss recovery. Han et al. [26] employed a mask-based self-learning approach with fine-tuning to mitigate the impact of packet loss. However, these methods heavily rely on dataset performance and require retraining in dynamic environments. Matrix recovery methods exploit the low-rank properties of measurement matrices and short-term motion patterns for packet loss recovery. Li et al. [27] introduced a TDoA-based localization method with low-rank approximation to handle random transmission delays and packet loss. This method eliminates the need for dataset construction and directly models the spatiotemporal correlations of target motion. However, the above studies do not account for NLoS ranging errors, which distort recovered features and degrade localization accuracy. Therefore, a robust solution that addresses both packet loss and NLoS errors is imperative for harsh environments.

To address the aforementioned challenges, this article proposes a ToF-based localization framework that integrates packet loss recovery and NLoS identification for accurate localization in harsh environments. The framework comprises a *continuous environment sensing module*, *packet loss recovery module*, *coarse localization module*, *NLoS identification module*, and *navigation refinement module*. With the iteration of localization and navigation, the continuous environment sensing module refines and updates dynamic environmental characteristics, i.e., ToF matrix, incorporating an NLoS table and the target's motion states (locations and velocity). Utilizing this ToF matrix, the packet loss recovery module generates completed ToFs via a rank-two matrix model. Coarse localization is estimated through ToFs' subsets, followed by NLoS identification using a historical residual trend analysis. Finally, refined navigation is performed utilizing the NLoS table. The proposed framework is evaluated on UWB-based hardware using DW1000 chips, deployed with six fixed anchors and one mobile tag for real-world experiments. The main contributions of this article are summarized below.

1) To the best of our knowledge, we propose the first framework that integrates packet loss recovery and NLoS identification, enabling continuous environment sensing for ToF-based localization in dynamic, harsh environments.
2) Utilizing spatiotemporal continuity constraints, a continuous environment sensing module is engineered to incorporate the historical NLoS table with the target's motion states. As localization iterates, the module continuously refines and updates environmental characteristics, such as the ToF matrix, thus yielding better packet loss recovery and NLoS identification.
3) Employing the dynamic environmental characteristics, a packet loss recovery module is designed using a rank-two matrix model to capture the target's motion state subspace, enabling packet loss recovery.
4) Leveraging the continuous environment sensing ability of the framework, we propose a historical residual trend analysis-based NLoS identification algorithm with an environmentally adaptive residual threshold strategy to enhance the accuracy of NLoS identification.

Throughout this article, a, $\mathbf{a}$, and $\mathbf{A}$ stand for a scalar, a column vector, and a matrix, respectively. $A[i, j]$ denotes the (i, j)th entry of $\mathbf{A}$. $\mathbb{R}^k$ and $\mathbb{R}^{m\times n}$ represent the domains of k-dimensional real vector and $m \times n$ real matrix, respectively. The ℓ_p norm, nuclear norm, and cardinality of a vector (or

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matrix) are denoted by $\|\cdot\|_p$, $\|\cdot\|_*$, and $|\cdot|$, respectively. The rest of this article is structured as follows: Section II introduces the system model. Section III presents the continuous environment sensing localization framework, and Section IV analyzes the experimental results. Finally, we draw the conclusions in Section V.

II. System Model

A. Overview of ToF-Based Localization

We consider an indoor localization scenario where a mobile tag (hereafter referred to as the target) moves freely within a 2-D space $\mathcal{L} \subset \mathbb{R}^2$, covered by a radio network of $N \geq 3$ fixed anchors, where N denotes the number of all anchors. For clarity, ToF measurements are treated as ranging measurements. The measured distance r_k^n between an anchor and the target can be modeled as

$$r_k^n = \left\|\mathbf{x}_k - \mathbf{x}^n\right\|_2 + e_k^n \tag{1}$$

$$e_k^n = \varepsilon_k^n + \kappa^n + \psi^n \tag{2}$$

where $\mathbf{x}_k$ and $\mathbf{x}^n$ denote the target's location at time k and the nth anchor's location, respectively. Here, the ranging error e_k^n consists of three components [22].

1) *Ranging offset* ε_k^n: Environment/location-dependent, arising from multipath or/and NLoS propagation.
2) *Systematic bias* κ^n: A hardware-dependent bias calibratable in advance.
3) *Measurement noise* ψ^n: Follows a zero-mean Gaussian distribution $\psi^n \sim \mathcal{N}(0, (\sigma_\psi^n)^2)$.

To improve localization accuracy, it is important to mitigate the impact of ranging error e_k^n. Particularly, the ranging offset ε_k^n is highly dynamic and unpredictable, influenced by the propagation environment (e.g., LoS/NLoS) and the material properties of NLoS obstructions.

ToF-based localization algorithms consist of nonlinear and linear methods. Newton–Raphson (NR) [28] and linear least squares (LLSs) [29] are classical nonlinear and linear methods, respectively. LLS has lower computational complexity, while NR is more robust in complex noise environments.

B. Problem Formulation

ToF-based localization involves two main steps. First, anchors and the target exchange packets using two-way ranging technology. Then, the timestamps of these packets are used to calculate ToFs and estimate the target's location. Fig. 1(a) illustrates ToF-based localization and navigation in a dynamic, harsh environment. As the target moves, the LoS/NLoS conditions change depending on the relative locations of the target, anchors, and obstructions. Packet loss is influenced by obstructions and further affected by signal collisions. When more targets participate in localization and navigation, the system must handle high-concurrency transmission and a higher ranging communication load, increasing the probability of signal collisions. Fig. 1(b) and (c) depict the target under varying packet loss and LoS/NLoS conditions along the same trajectory. Thus, even in the same environment, both ranging errors and packet loss probabilities may dynamically fluctuate. However, it is worth noting that the target's movement is continuous, with smooth forward or backward transitions. If we can establish the relationship between ToF frames in the ToF table, it becomes feasible to recover lost ToF data.

To address packet loss, matrix recovery techniques are employed to reconstruct a low-rank matrix from a subset of its entries, requiring only partial data. Let $\mathbf{A} \in \mathbb{R}^{m\times n}$ represents a rank-r noisy and incomplete data matrix with $r \ll \min\{m, n\}$. Define the binary mask matrix $\mathbf{M}_\Omega \in \{0, 1\}^{m\times n}$ as

$$M_\Omega[i, j] = \begin{cases} 1, & (i, j) \in \Omega \\ 0, & \text{otherwise} \end{cases} \tag{3}$$

where Ω denotes the set of observed entry indices.

In the absence of noise, the original matrix $\mathbf{A}$ can be accurately recovered by solving the following optimization problem:

$$\begin{aligned} &\min \|\mathbf{W}\|_* \\ &\text{s.t. } \mathbf{M}_\Omega \odot \mathbf{W} = \mathbf{M}_\Omega \odot \mathbf{A} \end{aligned} \tag{4}$$

where $\mathbf{W}$ is the completed matrix, and $\odot$ denotes the Hadamard (element-wise) product.

However, most existing packet loss recovery methods assume the signal is affected by zero-mean Gaussian white noise [27], overlooking the detection and suppression of positive bias NLoS errors. Current models, whether for packet loss recovery or NLoS identification, do not effectively address the combined effects of both on localization accuracy.

III. Proposed Framework and Algorithms

In this section, we present the proposed ToF-based localization framework for harsh environments, which enables packet loss recovery and NLoS identification as the localization and navigation process is continuously refined and updated. We then elaborate on the algorithms in the main modules.

A. System Framework and Workflow

The proposed framework comprises the following modules: *continuous environment sensing module*, *packet loss recovery module*, *coarse localization module*, *NLoS identification module*, and *navigation refinement module*, as shown in Fig. 2.

1) Continuous Environment Sensing Module: This module comprises an environment sensing database and a ToF matrix engine. It continuously collects data from other modules. Specifically, at time k, its inputs include the raw ToFs $\mathbf{r}_k$, estimated ToFs $\hat{\mathbf{r}}_{k-1}$, completed ToFs $\tilde{\mathbf{r}}_{k-1}$, and NLoS table $\mathbf{M}_{\Omega_{\text{nlos}}}$. The output of the module is a ToF matrix $\hat{\mathbf{A}}_k$.

2) Packet Loss Recovery Module: This module takes the raw ToFs $\mathbf{r}_k$ and the ToF matrix $\hat{\mathbf{A}}_k$ as inputs. It employs rank-r singular value decomposition (SVD), weight estimation, and matrix recovery to reconstruct missing ToFs, producing the completed ToFs $\tilde{\mathbf{r}}_k$.

3) Coarse Localization Module: Using the completed ToFs $\tilde{\mathbf{r}}_k$ as input, this module generates feasible ToF subsets S, which are then used to estimate the target's location $\{\tilde{\mathbf{x}}_{k,s}\}$, $s \in S$ via the LLS algorithm.

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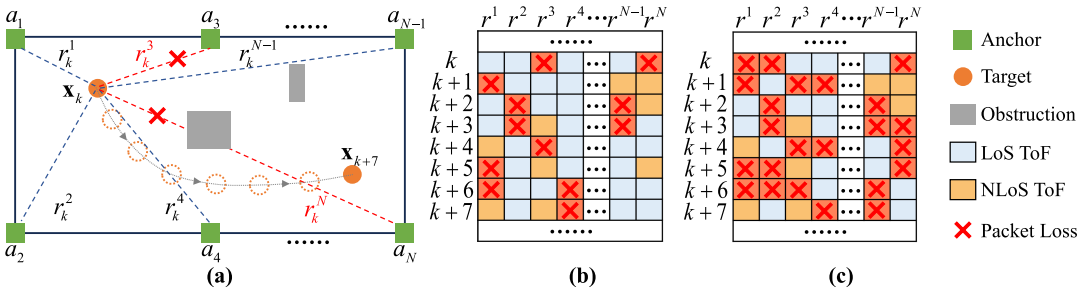


Fig. 1. ToF-based location and navigation in a dynamic harsh environment. (a) Localization and navigation scenario. (b) ToF table with low packet loss rate. (c) ToF table with high packet loss rate.

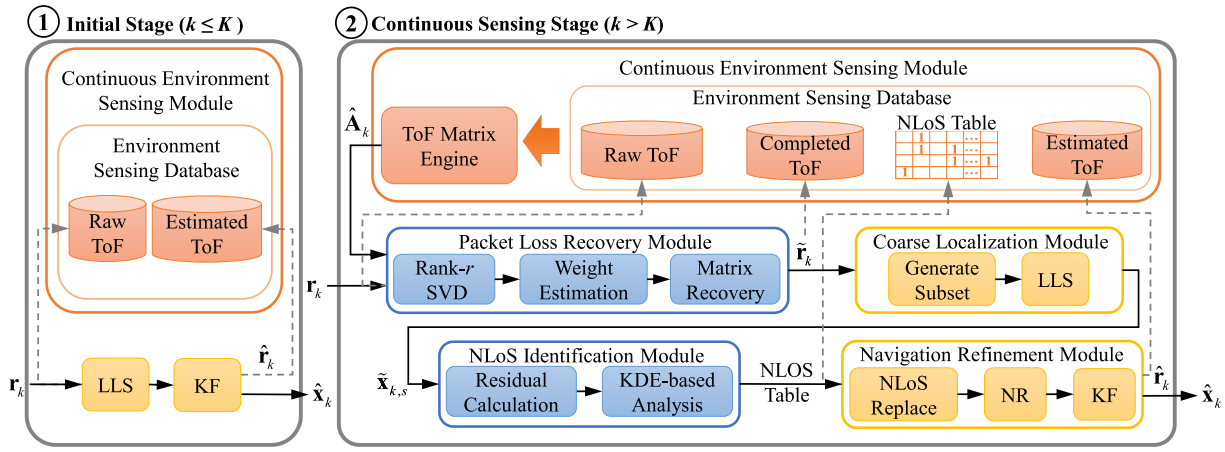


Fig. 2. Architecture of the proposed framework. Solid arrows represent the algorithm flow and dotted arrows represent online data acquisition.

4) NLoS Identification Module: This module takes the location estimates $\{\tilde{\mathbf{x}}_{k,s}\}, s \in S$ as input and computes the ToF residuals for each anchor. Leveraging the historical residual trend, it employs KDE to assess NLoS conditions and outputs the NLoS table $\mathbf{M}_{\Omega_{\text{nlos}}}$.

5) Navigation Refinement Module: Utilizing the NLoS table $\mathbf{M}_{\Omega_{\text{nlos}}}$ as input, this module replaces NLoS ToFs with those derived from coarse location estimates. It then refines the location through the NR method and Kalman Filter (KF), yielding the refined location $\hat{\mathbf{r}}_k$.

B. Continuous Environment Sensing Module

The continuous environment sensing module consists of an environment sensing database and a ToF matrix engine. Initially, due to a lack of sufficient data to generate the ToF matrix $\hat{\mathbf{A}}_k$, the module operates in two stages: the initial stage and the continuous sensing stage. In the initial stage, when data are insufficient ($k \le K$), only the LLS and KF algorithms are used for localization, and the estimated ToFs $\hat{\mathbf{r}}_k$ are stored in the environment sensing database. The time window size K is determined by the target's motion speed and the ranging frequency: a larger K can be adopted when the target moves slowly or the ranging frequency is high, ensuring higher similarity between historical and current ToFs; otherwise, K should be smaller to avoid nonlinearity caused by rapid changes in motion. Once data is sufficient ($k > K$), the continuous sensing stage is performed. The database continues to store the outputs from the other four

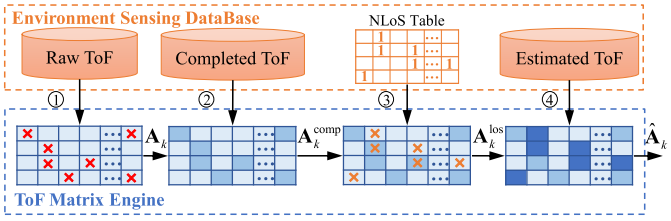


Fig. 3. Workflow of the ToF matrix engine.

modules. With the historical raw ToFs $\mathbf{r}_i$, historical estimated ToFs $\hat{\mathbf{r}}_i$, historical completed ToFs $\tilde{\mathbf{r}}_i$, $i = k-K, \ldots, k-1$, and NLoS table $\mathbf{M}_{\Omega_{\text{nlos}}}$, ToF matrix $\hat{\mathbf{A}}_k$ can be constructed by the ToF matrix engine. As localization progresses, the module acquires more environmental characteristics stored in the environment sensing database, gradually enhancing the performance of the ToF matrix and reducing errors caused by earlier instability in NLoS identification and ToF recovery.

Since the dimension of matrix $\hat{\mathbf{A}}_k$ affects the recovery performance, we set $\hat{\mathbf{A}}_k \in \mathbb{R}^{N\times K}$, where K represents the size of the time window, i.e., K frames of historical data are required to construct matrix $\hat{\mathbf{A}}_k$. To make the most of the information in the environment sensing database, the workflow of the ToF matrix engine is shown in Fig. 3 and the workflow to generate ToF matrix $\hat{\mathbf{A}}_k$ are outlined as follows:

1) First, the historical raw ToFs $\mathbf{r}_i$, $i = k-K, \ldots, k-1$ is used to construct the ToF matrix $\mathbf{A}_k \in \mathbb{R}^{N\times K}$ as

$$\mathbf{A}_k = \left[\mathbf{r}_{k-K}, \ldots, \mathbf{r}_{k-1}\right] \tag{5}$$

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Algorithm 1 Continuous Environment Sensing

Input: Raw ToFs $\mathbf{r}_k$, estimated ToFs $\hat{\mathbf{r}}_{k-1}$, completed ToFs $\tilde{\mathbf{r}}_{k-1}$, NLoS table $\mathbf{M}_{\Omega_{\text{nlos}}}$
; **Output:** Estimated ToF matrix $\hat{\mathbf{A}}_k$;
while $k \leq K$ **do**
  Collect $\mathbf{r}_k$, $\hat{\mathbf{r}}_{k-1}$;
end
Collect $\mathbf{r}_k$, $\hat{\mathbf{r}}_{k-1}$, $\tilde{\mathbf{r}}_{k-1}$, and $\mathbf{M}_{\Omega_{\text{nlos}}}$;
Construct $\mathbf{A}_k$ using Eq. (5);
Calculate and obtain $\mathbf{A}_k^{\text{comp}}$using Eq. (6);
Calculate and obtain $\mathbf{A}_k^{\text{los}}$ using Eq. (7);
Calculate and obtain $\hat{\mathbf{A}}_k$ using Eq. (8);
Return $\hat{\mathbf{A}}_k$.

where $\mathbf{r}_k = [r_k^1, \ldots, r_k^N]^{\mathrm{T}}$ denotes the raw ToFs at time k. As shown by arrow ① in Fig. 3, the matrix $\mathbf{A}_k$ contains several red crosses, indicating missing data at the corresponding positions.

2) Next, utilizing the historical completed ToFs $\tilde{\mathbf{r}}_i$, $i = k-K, \ldots, k-1$, construct the historical completed ToF matrix $\mathbf{B}_k = \left[\tilde{\mathbf{r}}_{k-K}, \ldots, \tilde{\mathbf{r}}_{k-1}\right]$ as shown in Fig. 3 arrow ②. Then, we substitute $\mathbf{B}_k$ into $\mathbf{A}_k$ to obtain the current completed ToF matrix $\mathbf{A}_k^{\text{comp}}$ as

$$\mathbf{A}_k^{\text{comp}} = \begin{cases} A_k\left[i, j\right], & (i, j) \in \Omega \\ B_k\left[i, j\right], & \text{otherwise.} \end{cases} \tag{6}$$

3) Then, define the NLoS table $M_{\Omega_{\text{nlos}}}[i, j] = 1$ when $(i, j) \in \Omega_{\text{nlos}}$, otherwise $M_{\Omega_{\text{nlos}}}[i, j] = 0$. Utilize $\mathbf{M}_{\Omega_{\text{nlos}}}$ to delete the NLoS ToFs, as shown by the orange crosses from the NLoS table in Fig. 3, and LoS ToF matrix $\mathbf{A}_k^{\text{los}}$ is obtained as

$$\mathbf{A}_k^{\text{los}} = \mathbf{A}_k^{\text{comp}} - \mathbf{M}_{\Omega_{\text{nlos}}} \odot \mathbf{A}_k^{\text{comp}}. \tag{7}$$

4) Subsequently, using the historical estimated ToFs $\hat{\mathbf{r}}_i$, $i = k-K, \ldots, k-1$, define $\mathbf{C}_k = \left[\hat{\mathbf{r}}_{k-K}, \ldots, \hat{\mathbf{r}}_{k-1}\right]$ as shown in Fig. 3 arrow ④, which can be filled in the gaps deleted by the NLoS table, and $\hat{\mathbf{A}}_k$ is obtained as

$$\hat{\mathbf{A}}_k = \mathbf{A}_k^{\text{los}} + \mathbf{M}_{\Omega_{\text{nlos}}} \odot \mathbf{C}_k. \tag{8}$$

Finally, the ToF matrix $\hat{\mathbf{A}}_k$ is obtained.

The continuous environment sensing algorithm is summarized in Algorithm 1.

C. Packet Loss Recovery Module

To leverage the spatio-temporal correlation of the target's movements, we introduce the application of the GROUSE [30] in packet loss recovery. At time k, there exists a ToF matrix $\hat{\mathbf{A}}_k$ that can be performed by SVD as

$$\hat{\mathbf{A}}_k = \hat{\mathbf{U}}_k \hat{\Sigma}_k \hat{\mathbf{V}}_k^{\mathrm{T}} \tag{9}$$

where $\hat{\mathbf{U}}_k \in \mathbb{R}^{N\times N}$, $\hat{\mathbf{V}}_k \in \mathbb{R}^{K\times K}$ denote unitary matrices, and $\hat{\Sigma}_k$ contains the singular values of $\hat{\mathbf{A}}_k$. By applying the rank-r decomposition method, matrix $\hat{\mathbf{A}}_{k,r}$ is obtained by

$$\hat{\mathbf{A}}_{k,r} = \hat{\mathbf{U}}_{k,r} \hat{\Sigma}_{k,r} \hat{\mathbf{V}}_k^{\mathrm{T}} = \hat{\mathbf{U}}_{k,r} \left(\hat{\Sigma}_{k,r} \hat{\mathbf{V}}_k^{\mathrm{T}}\right) = \mathbf{U}_k \mathbf{R}_k^{\mathrm{T}} \tag{10}$$

Algorithm 2 GROUSE for Packet Loss Recovery

Input: ToF matrix $\hat{\mathbf{A}}_k$, current raw ToFs $\mathbf{r}_k$;
Output: Completed ToFs $\tilde{\mathbf{r}}_k$;
According to $\mathbf{r}_k$, obtain Ω_k;
Calculate rank-r SVD of $\hat{\mathbf{A}}_k$, and obtain $\mathbf{U}_k, \mathbf{R}_k$ using Eqs. (9) and (10);
Calculate weight using Eq. (11);
Obtain $\tilde{\mathbf{r}}_k$ using Eq. (12);
Return $\tilde{\mathbf{r}}_k$.

where $\hat{\mathbf{U}}_{k,r}$ is represented by the first r column of $\hat{\mathbf{U}}_k$; $\hat{\Sigma}_{k,r}$ is represented by the first r singular values of $\hat{\Sigma}_k$; $\mathbf{U}_k$ and $\mathbf{R}_k$ denote orthogonal and symmetric matrices, respectively.

Consider a new ToF vector $\mathbf{r}_k$ with missing data, where the observed entries are denoted by $\Omega_k \subseteq \{1, \ldots, N\}$. Under the observed entries Ω_k, the weight coefficient $\mathbf{w}_k$ can be defined by the least square method, i.e.,

$$\hat{\mathbf{w}}_k = \arg\min_{\mathbf{w}} \left\| \mathbf{U}_{\Omega_k} \mathbf{w} - \mathbf{r}_{\Omega_k} \right\|_2^2. \tag{11}$$

Given that the cardinality of Ω_k less than 2, even a small amount of noise will lead to significant errors in the weight estimation of (11), which in turn results in substantial inaccuracies in the completed ToFs. To address this issue, we utilize the last estimated ToFs to recover from packet loss. The completed ToF vector $\tilde{\mathbf{r}}_k$ is obtained by

$$\tilde{\mathbf{r}}_k = \begin{cases} r_{\Omega_k}\left[i\right], & i \in \Omega_k \\ \hat{r}_{k-1}\left[i\right], & i \notin \Omega_k, |\Omega_k| \leq 2 \\ \mathbf{U}_k \hat{\mathbf{w}}_k, & \text{otherwise.} \end{cases} \tag{12}$$

The detailed algorithm for packet loss recovery is shown in Algorithm 2. Notably, we have not analyzed the value of r. Actually, as the target's motion changes, the rank of $\hat{\mathbf{A}}_k$ will vary. Initially, under static conditions, we consider applying column elementary transformations to the ToF matrix as

$$\hat{\mathbf{A}}_k \sim \begin{bmatrix} d_{k-K}^1 & 0 & \cdots & 0 \\ d_{k-K}^2 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ d_{k-K}^N & 0 & \cdots & 0 \end{bmatrix}. \tag{13}$$

In the static state, the ideal ToF matrix satisfies $d_k^n = d_{k-i}^n$, $(i = 1, \ldots, K, n = 1, \ldots, N)$, making $\hat{\mathbf{A}}_k$ a rank-one matrix.

Under dynamic conditions, the target may exhibit complex motion patterns, such as varying speed. However, it can be assumed that within a short time window, specifically within the matrix sliding window K, the target moves in uniform linear motion. The column elementary transformation of the ToF matrix can be performed as

$$\hat{\mathbf{A}}_k \sim \begin{bmatrix} d_{k-K}^1 & \Delta_k^1 & \cdots & 0 \\ d_{k-K}^2 & \Delta_k^2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ d_{k-K}^N & \Delta_k^N & \cdots & 0 \end{bmatrix} \tag{14}$$

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where $\Delta_i^n = d_i^n - d_{i-1}^n$, $(i = k-K+1, \ldots, k-1,\ n = 1, \ldots, N)$. Consider that in a short time slice, $\Delta_{k-K+1}^n = \Delta_{k-K+2}^n = \cdots = \Delta_{k-1}^n$, $(n = 1, \ldots, N)$.

Algorithm 2 presents the packet loss recovery algorithm. Given that (13) is a special form of (14), we set $r = 2$ in Algorithm 2.

D. Coarse Localization Module and NLoS Identification Module

For 2-D localization, at least 3 anchors are required to solve the localization equations using the LLS algorithm [29]. If the number of anchors $N \geq 3$, subsets of N_{sub} anchors can be formed, where $3 \leq N_{\text{sub}} \leq N$. Subsets with anchors nearly aligned should be excluded. Then, a subset $s \in S$ can compute a location estimate $\tilde{\mathbf{x}}_{k,s}$, where S denotes the set of all feasible subsets. Subsequently, according to the location of anchor a_n, the coarse estimated ToF $r_{k,s}^n$ can be calculated as

$$r_{k,s}^n = \left\| \tilde{\mathbf{x}}_{k,s} - \mathbf{x}^n \right\|_2. \tag{15}$$

The residual of anchor a_n for the subset s is defined as

$$\gamma_{k,s}^n = \left\| r_k^n - r_{k,s}^n \right\|_1. \tag{16}$$

According to the residuals from all subsets, the anchor's average residual is obtained by

$$\lambda_{k,n} = \frac{1}{|\{s \in S \mid a_n \in s\}|} \sum_{\{s \in S | a_n \in s\}} \gamma_{k,s}^n. \tag{17}$$

In practical ToF-based systems, NLoS propagation introduces positive bias errors because signals travel along longer indirect paths. Consequently, NLoS anchors generally exhibit larger average residuals than LoS anchors. Leveraging this property, we reformulate the NLoS identification problem as a 1D clustering task on the average residuals, enabling adaptive handling of multiple NLoS issues.

To implement this 1-D clustering, we employ the KDE method to estimate the distribution of the average residuals. For a series of independent and identically distributed samples $\{\lambda_{k,1}, \ldots, \lambda_{k,N}\}$, the KDE is expressed as [21]

$$\hat{f}_{k,h}(\lambda) = \frac{1}{N} \sum_{i=1}^{N} Q\left(\frac{\lambda - \lambda_{k,i}}{h} \right) \tag{18}$$

where $Q(\cdot)$ is a nonnegative function called the kernel (we used a Gaussian kernel in this work), and h controls the smoothness of the fit distribution. Based on the distribution, we perform the following procedure to identify NLoS ToFs.

1) First, the mean μ_k and standard deviation σ_k of the residuals are calculated.
2) The M_{nlos} historical residual means $\{\mu_i\}_{i=k-1}^{k-M_{\text{nlos}}}$ and standard deviation $\{\sigma_i\}_{i=k-1}^{k-M_{\text{nlos}}}$ are used to calculate the mean threshold μ_{thr} and standard deviation threshold σ_{thr}, respectively. The threshold is greater than ξ percentage of elements in the historical data

$$\mu_{\text{thr}} \geq \mu_{(\lceil \xi M_{\text{nlos}} \rceil)} \tag{19}$$

$$\sigma_{\text{thr}} \geq \sigma_{(\lceil \xi M_{\text{nlos}} \rceil)} \tag{20}$$

where $\mu_{(\lceil \xi M_{\text{nlos}} \rceil)}$ represents the (ξM_{nlos})th smallest element in the set $\{\mu_i\}_{i=k-1}^{k-M_{\text{nlos}}}$, and $\lceil \cdot \rceil$ represents the integer up function; likewise for $\sigma_{(\lceil \xi M_{\text{nlos}} \rceil)}$.

3) If $\mu_k > \mu_{\text{thr}}$ and $\sigma_k > \sigma_{\text{thr}}$, NLoS identification is performed in step 4); otherwise, NLoS identification is not performed, in step 6).
4) According to the distribution $\hat{f}_{k,h}(\lambda)$, the local extrema (local maxima and local minima) are calculated. According to the number of local extrema, the data can be clustered, i.e., the residual values contained in each peak correspond to a cluster of anchors.
5) If there is only one clustering result, all of them are regarded as LoS. If there are two or more clusters, the anchors in the cluster with the largest overall value are considered NLoS.
6) Finally, the output is the NLoS table.

The KDE-based NLoS Identification Algorithm via ToF residual analysis is summarized in Algorithm 3.

Algorithm 3 KDE-Based ToF Residual Analysis for NLoS Identification

Input: Coarse location estimates $\tilde{\mathbf{x}}_{k,s}$; threshold coefficient ξ; smoothing parameter h;
Output: NLoS table $\mathbf{M}_{\Omega_{\text{nlos}}}$;

```
Calculate γ^n_{k,s} using Eqs. (16) and (17);
Calculate μ_k and σ_k of the current residuals;
Calculate μ_thr and σ_thr using Eqs. (19) and (20);
if μ_k > μ_thr and σ_k > σ_thr then
    Calculate f̂_{k,h}(λ) using Eq. (18);
    Calculate the local extremum values of f̂_{k,h}(λ);
    Classify residuals according to the number of the
     local extrema;
    if Number of clusters≥ 2 then
        Determine that certain anchors are in NLoS;
    else
        Determine that all anchors are in LoS;
    end
else
    Determine that all anchors are in LoS;
end
Return M_{Ω_nlos}.
```

E. Navigation Refinement Module

Based on the NLoS table obtained from KDE, we mitigate the impact of NLoS by replacing NLoS ToFs with the estimated ToFs computed from (15). Using the NR method [28], the location estimate $\hat{\mathbf{x}}_k^{\text{NR}}$ is obtained with a convergence threshold ζ and a maximum number of iterations M_{NR}.

To further smooth the localization trajectory and suppress noisy observations under harsh environments, a KF is applied. The system state at time k is defined as $\mathbf{x}_{k|k} = [\hat{\mathbf{x}}_k^{\text{T}}, \hat{\mathbf{v}}_k^{\text{T}}]^{\text{T}} \in \mathbb{R}^4$, where $\hat{\mathbf{x}}_k$ and $\hat{\mathbf{v}}_k$ denote the filtered location and velocity, respectively. Without inertial measurements, the velocity is approximated by consecutive location differences, i.e. $\hat{\mathbf{v}}_k = \hat{\mathbf{x}}_k - (\hat{\mathbf{x}}_{k-1})/\Delta t$, where Δt is the sampling interval.

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The state transition model is given by

$$\mathbf{x}_{k|k-1} = \mathbf{F}\,\mathbf{x}_{k-1|k-1}, \quad \mathbf{F} = \begin{bmatrix} 1 & 0 & \Delta t & 0 \\ 0 & 1 & 0 & \Delta t \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \tag{21}$$

with process noise covariance

$$\mathbf{Q} = \sigma_a^2 \begin{bmatrix} \frac{1}{4}\Delta t^4 & 0 & \frac{1}{2}\Delta t^3 & 0 \\ 0 & \frac{1}{4}\Delta t^4 & 0 & \frac{1}{2}\Delta t^3 \\ \frac{1}{2}\Delta t^3 & 0 & \Delta t^2 & 0 \\ 0 & \frac{1}{2}\Delta t^3 & 0 & \Delta t^2 \end{bmatrix} \tag{22}$$

where $\mathbf{F}$ is the state transition matrix, $\mathbf{x}_{k|k-1}$ the predicted state, and σ_a^2 the process acceleration noise variance.

The KF takes the NR location estimate as the observation, i.e., $\mathbf{z}_k = \hat{\mathbf{x}}_k^{\mathrm{NR}} \in \mathbb{R}^2$. The observation model is defined as

$$\mathbf{z}_k = \mathbf{H}\,\mathbf{x}_k + \mathbf{w}_k, \quad \mathbf{H} = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{bmatrix} \tag{23}$$

where $\mathbf{H}$ is the observation matrix and $\mathbf{w}_k \sim \mathcal{N}(\mathbf{0}, \mathbf{R}_k)$ is the observation noise.

To account for NLoS effects, the observation covariance $\mathbf{R}_k$ is adaptively obtained from the NLoS table via geometry-propagation [31]. Linearizing at $\hat{\mathbf{x}}_k^{\mathrm{NR}}$, the unit LoS vector to anchor a_n is

$$\mathbf{u}_{k,n} = \frac{\hat{\mathbf{x}}_k^{\mathrm{NR}} - \mathbf{x}^n}{\left\|\hat{\mathbf{x}}_k^{\mathrm{NR}} - \mathbf{x}^n\right\|_2}. \tag{24}$$

Stacking these row vectors yields the geometry matrix $\mathbf{G}_k = [\mathbf{u}_{k,1}^{\mathrm{T}}, \ldots, \mathbf{u}_{k,N}^{\mathrm{T}}]^{\mathrm{T}}$. From the NLoS table, each anchor a_n is assigned a range variance $\sigma_{r,n}^2$ based on its average residual $\lambda_{k,n}$ from the NLoS identification stage

$$\sigma_{r,n}^2 = \sigma_{\mathrm{los}}^2 \left[1 + \alpha \left(\frac{\lambda_{k,n}}{\mathrm{median}_{\mathrm{hist}}\,(\lambda_n)}\right)^p\right] \tag{25}$$

where σ_{los}^2 is the nominal LoS range variance, $\mathrm{median}_{\mathrm{hist}}(\lambda_n)$ is the historical median of the residuals over a sliding window M_{nlos}, $\alpha > 0$ is an inflation factor controlling NLoS penalization, and p adjusts the sensitivity to residual growth. The range variance matrix is then

$$\mathbf{\Sigma}_{r,k} = \mathrm{diag}\left(\sigma_{r,1}^2, \ldots, \sigma_{r,N}^2\right). \tag{26}$$

The observation covariance $\mathbf{R}_k$ is given by

$$\mathbf{R}_k = \tau \left(\mathbf{G}_k^{\mathrm{T}} \mathbf{\Sigma}_{r,k}^{-1} \mathbf{G}_k\right)^{-1} \tag{27}$$

where $\tau \geq 1$ is a robustness inflation factor.

The KF prediction–update equations are given by

$$\text{(Prediction)} \quad \mathbf{x}_{k|k-1} = \mathbf{F}\,\mathbf{x}_{k-1|k-1} \tag{28}$$

$$\mathbf{P}_{k|k-1} = \mathbf{F}\,\mathbf{P}_{k-1|k-1}\,\mathbf{F}^{\mathrm{T}} + \mathbf{Q} \tag{29}$$

$$\text{(Update)} \quad \mathbf{K}_k = \mathbf{P}_{k|k-1}\mathbf{H}^{\mathrm{T}} \left(\mathbf{H}\mathbf{P}_{k|k-1}\mathbf{H}^{\mathrm{T}} + \mathbf{R}_k\right)^{-1} \tag{30}$$

$$\mathbf{x}_{k|k} = \mathbf{x}_{k|k-1} + \mathbf{K}_k \left(\mathbf{z}_k - \mathbf{H}\,\mathbf{x}_{k|k-1}\right) \tag{31}$$

$$\mathbf{P}_{k|k} = (\mathbf{I}_4 - \mathbf{K}_k\,\mathbf{H})\,\mathbf{P}_{k|k-1} \tag{32}$$

where $\mathbf{P}_{k|k-1}$ and $\mathbf{P}_{k|k}$ are the predicted and updated covariance matrices; $\mathbf{K}_k$ is the Kalman gain; and $\mathbf{I}_4$ is the 4×4 identity matrix. The filtered location $\hat{\mathbf{x}}_k$ from the updated state $\mathbf{x}_{k|k}$ is then used as the final localization output at time k.

F. Computational Complexity Analysis

This section analyzes the computational complexity of the proposed framework, which consists of five main modules. The continuous environment sensing module primarily constructs the ToF matrix with minimal computation. The packet loss recovery module employs SVD and weight estimation, both with a complexity of $\mathcal{O}(N^2K)$, where N is the number of anchors and K is the time window. The coarse localization module includes ToF subset generation and LLS-based localization. Since subset generation involves selecting N_{sub} ToFs from N, the number of possible subsets is $S = \binom{N}{N_{\mathrm{sub}}}$, resulting in a selection complexity of $\mathcal{O}(S)$. Each subset requires LLS estimation with a per-subset complexity of $\mathcal{O}(N_{\mathrm{sub}}^3)$, leading to an overall complexity of $\mathcal{O}(SN_{\mathrm{sub}}^3)$. To maintain computational feasibility, N_{sub} is typically restricted to 3 or 4 when $N > 4$. For NLoS identification, standard KDE has a complexity of $\mathcal{O}(S^2)$ due to pairwise kernel evaluations. Optimized methods, such as KD-tree-based approaches [32], reduce this to $\mathcal{O}(S \log S)$, improving efficiency for large S. The navigation refinement module, which involves NR and KF, has a complexity of $\mathcal{O}(N^3)$ due to matrix inversion and nonlinear optimization, making it the most computationally intensive step. The total complexity of the framework is

$$\mathcal{O}\left(N^2K\right) + \mathcal{O}\left(SN_{\mathrm{sub}}^3\right) + \mathcal{O}\left(S \log S\right) + \mathcal{O}\left(N^3\right). \tag{33}$$

Despite optimizations in KDE and LLS, subset generation and navigation refinement remain the primary computational bottlenecks. The combinatorial growth of S leads to significant computational costs, even with N_{sub} restricted to 3 or 4. Additionally, navigation refinement involves high-cost matrix operations with $\mathcal{O}(N^3)$ complexity. Given practical constraints on anchor placement and communication range, the number of accessible anchors is typically limited to 10. Increasing the number of anchors would not only raise computational overhead, particularly in subset selection and LLS calculations, but also introduce challenges in network synchronization and data transmission. The proposed framework has been validated in a controlled environment with six anchors, demonstrating its feasibility for real-time localization applications.

IV. Experimental Results

In this section, we present the experimental results conducted within a harsh indoor environment to validate the effectiveness of the proposed framework.

A. Experimental Setup

1) Scenario: We conduct real-world experiments in an indoor office environment (16.60 × 8.80 m) to assess the applicability of our framework. As shown in Fig. 4(a), the scene includes two pillars with six anchors. The anchors are fixed at a height of 2.4 ± 0.1 m, as shown in Fig. 4(b). Additionally, the target tag, placed at a height of 1.4 m, is connected to a personal computer via a serial cable for real-time data transmission, as depicted in Fig 4(c).

2) Hardware: We implement the proposed framework using commodity UWB devices (UWB anchor and tag), as shown in Fig. 4(b). The devices are equipped with a DW1000 chip,

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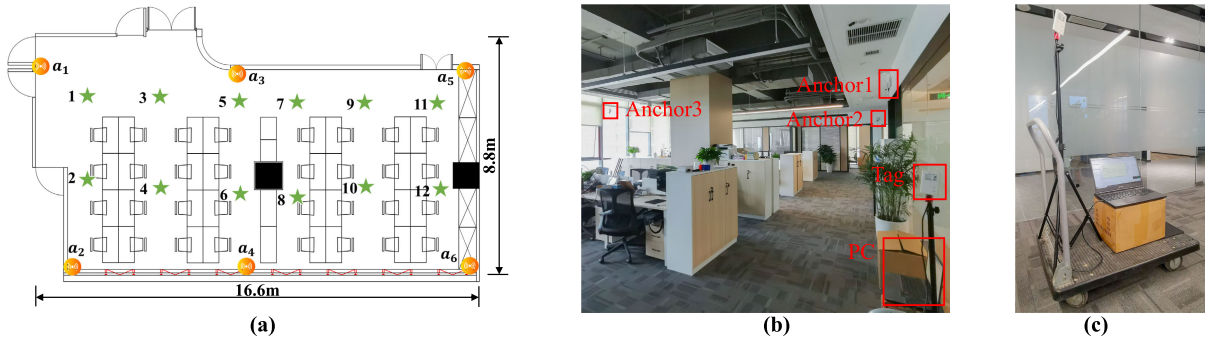


Fig. 4. Experiment scenarios and test platform. (a) black solid rectangles represent the pillars, the orange icons denote the fixed UWB anchors, and the green stars indicate the designated locations for the localization experiments. (a) 2-D top view. (b) Experiment environment. (c) Test platform.

configured with a 499.2 MHz bandwidth, 3993.6 MHz carrier frequency, 6.8 Mb/s data rate, and 64 MHz pulse repetition frequency.

3) Datasets: As shown in Table I, data were collected at approximate packet loss rates of 10%, 30%, and 50%, with 400 samples gathered at the green star locations shown in Fig. 4(a). The ground-truth locations of the target were obtained using SNDWAY's SW-M50 laser rangefinder. For the navigation experiment, a cart followed a predefined trajectory in the office, and data were collected under packet loss rates of 10%, 20%, and 30%. The different packet loss rates were primarily achieved by varying the number of target tags in the environment. Specifically, packet loss rates of 10%, 20%, 30%, and 50% correspond to configurations with 1, 7, 13, and 20 tags, respectively.

4) Comparative Methods: In the localization and navigation experiments, our proposed framework is compared with several state-of-the-art (SOTA) approaches for harsh environments. We compare the proposed framework with five other localization methods.

1) *Clustering-Based Trilateration (CBT) Method [33]:* This method improves trilateration using clustering to handle intersection uncertainties.
2) *Error-Summation-Incorporated Newton Iteration (ESINI) Method [34]:* This method uses a Taylor expansion to estimate the effects of decreasing random biases.
3) *Low-Rank Matrix Approximation (LRMA) Method [27]:* This method completes the rank-two ToF matrix by minimizing its nuclear norm.
4) *N-only Method [21]:* Only NLoS Identification via KDE-based residual analysis without packet loss recovery.
5) *R-Only Method [30]:* Only packet loss recovery via the GROUSE algorithm without NLoS identification.

5) Parameter Settings: In the experiment, the model hyperparameters are set as shown in Table II. For the KF tracking used in all comparative experiments, the initial target motion state is set to the location from the coarse localization of the first frame with zero velocity. The initial estimation error covariance matrix is set to $\mathbf{P}_0 = \text{diag}(0.5^2, 0.5^2, 0.2^2, 0.2^2)$, where location and velocity are in meters and m/s, respectively. The measurement noise covariance matrix is given by $\mathbf{R}_0 = \text{diag}(\sigma_{\text{los}}^2)$.

B. Localization Performance in Harsh Environments

Table I presents the localization performance metrics of the proposed framework across three datasets collected under different packet loss rates of approximately 10%, 30%, and 50% in a static scenario. The evaluation metrics include the Mean Absolute Error (MAE) and the 80th percentile MAE (MAE80). In Exp. 1 (10% packet loss), the proposed framework achieves MAE and MAE80 of 13.74 and 40.44 cm, respectively, outperforming CBT, ESINI, LRMA, N-only, and R-only in MAE80 by 36.16%, 39.27%, 42.45%, 49.20%, and 34.75%, respectively. As the packet loss rate increases, the proposed framework maintains strong performance. In Exp. 2 (30% packet loss), it achieved MAE and MAE80 of 17.71 and 43.37 cm, respectively. Even under 50% packet loss in Exp. 3, most localization errors remained within 50 cm, with MAE80 reduced by 64.97%, 50.18%, 55.30%, 42.68%, and 21.82% compared to CBT, ESINI, LRMA, N-only, and R-only, respectively. Overall, while other methods degraded markedly with increasing packet loss, the proposed framework demonstrated strong robustness.

Fig. 5(a) presents the Cumulative Distribution Function (cdf) of localization errors for the proposed framework compared to SOTA models in Exp. 1. Under a low packet loss rate of 10%, the proposed framework achieves the best cdf performance, surpassing CBT, ESINI, and LRMA. LRMA performance degrades to a level comparable with CBT and ESINI due to the combined effects of NLoS conditions and packet loss. In the N-only model, packet loss weakens NLoS identification, whereas the R-only model benefits from effective packet loss recovery, improving accuracy. Under a high packet loss rate of 50%, the proposed framework and R-only model markedly outperform others, underscoring the importance of the packet loss recovery module. However, the N-only model performs similar to ESINI, indicating limited benefit from NLoS identification alone. Integrating packet loss recovery with NLoS identification allows the proposed framework to surpass the R-only model, as recovered ToFs not only replace missing data but also improve NLoS detection. Fig. 5(c) further compares robustness under packet loss rates of approximately 10% and 50%. With 10% packet loss, all

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TABLE I
QUANTITATIVE EVALUATIONS OF PROPOSED FRAMEWORK WITH COMPARED METHODS IN DIFFERENT EXPERIMENTS. BEST RESULTS ARE HIGHLIGHTED, AND SECOND-BEST RESULTS ARE UNDERLINED

Exp.	State	Frames	Packet Loss Rate/%						Metrics	Methods/cm					
			a_1	a_2	a_3	a_4	a_5	a_6		CBT	ESINI	LRMA	N-only	R-only	**Prop.**
1	Static	4800	13.13	14.56	14.08	12.94	13.19	16.48	MAE	30.59	27.93	33.62	42.18	23.33	**13.74**
									MAE80	63.35	66.59	70.27	79.60	61.98	**40.44**
2	Static	4800	28.50	27.90	27.00	27.44	27.17	30.33	MAE	37.50	36.17	36.31	41.40	24.21	**17.71**
									MAE80	83.56	75.06	70.57	84.65	63.24	**43.37**
3	Static	4800	48.85	47.42	47.75	45.54	48.54	50.50	MAE	47.98	50.85	58.68	44.73	26.15	**24.58**
									MAE80	143.00	100.54	112.05	87.39	64.07	**50.09**
4	Mobile	1172	11.52	11.01	10.33	13.14	9.17	12.37	MAE	88.14	93.01	104.16	91.02	79.48	**45.96**
									MAE80	146.12	158.48	146.06	151.02	135.65	**87.64**
5	Mobile	1154	19.24	18.72	17.85	23.22	16.12	21.66	MAE	102.44	97.06	120.64	87.05	81.92	**47.42**
									MAE80	201.94	162.91	172.10	155.12	139.89	**95.01**
6	Mobile	1197	29.57	29.16	29.82	29.49	27.57	32.75	MAE	116.02	96.78	131.76	103.02	84.27	**52.06**
									MAE80	240.72	163.08	203.28	177.26	138.08	**111.36**

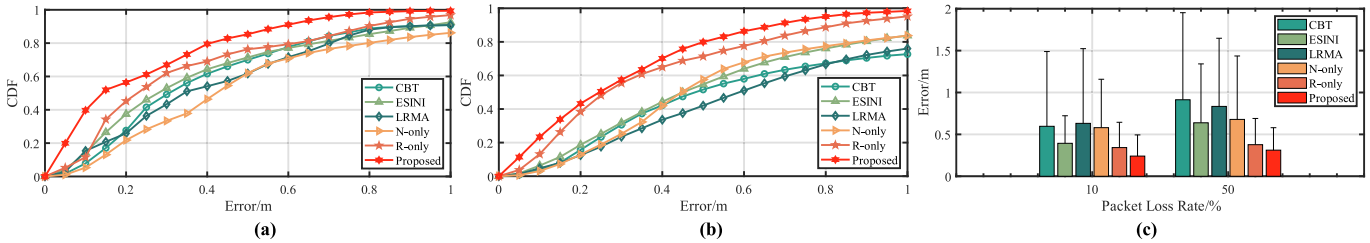


Fig. 5. Localization performance in harsh environments and the robustness against different packet loss rates. (a) CDF of the localization error in Exp. 1. (b) CDF of the localization error in Exp. 3. (c) Robustness against different packet loss rates.

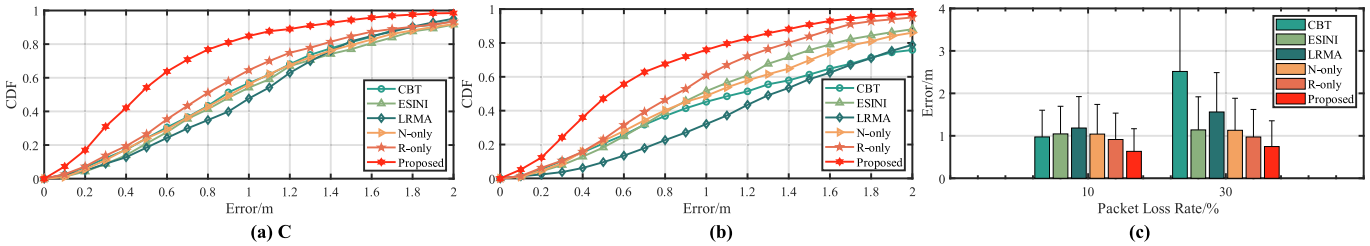


Fig. 6. Navigation performance in harsh environments and the robustness against different packet loss rates. (a) CDF of the localization error in Exp. 4. (b) CDF of the localization error in Exp. 6. (c) Robustness against different packet loss rates.

TABLE II
PARAMETER SETTINGS

Module	Parameter	Setting
Continuous Environment Sensing	# All anchors N	6
	# Time window K	4
NLoS identification	# Historical residuals M_{nlos}	100
	Threshold coefficient ξ	0.8
	Smoothing parameter h	0.01
Navigation Refinement	NR iterations M_{NR}	20
	NR convergence threshold ζ	0.001m
	Acceleration noise variance σ_a^2	$0.1\text{m}^2/\text{s}^4$
	Nominal LoS range variance σ_{los}^2	0.01m^2
	NLoS weighting params (α, p)	$(1, 1)$
	Inflation factor τ	1.2

methods achieve average localization errors within 0.6 m, though the proposed framework remains superior. In contrast, most methods degrade substantially at 50% packet loss, while the proposed framework sustains high accuracy.

C. Navigation Performance in Harsh Environments

To demonstrate the performance of the proposed framework in dynamic environments, we collected three mobile datasets under different packet loss rates, corresponding to Exps. 4–6 in Table I. The movement speeds in these three experiments were kept as similar as possible, with packet loss rates of approximately 10%, 20%, and 30%, respectively. In dynamic scenarios, a 50% packet loss rate was not tested because it often yields localization frames with only one or two valid ToF measurements, which severely hinders fair comparison with conventional methods.

In Exp. 4, as shown in Table I, the proposed framework achieves MAE and MAE80 of 45.96 and 87.64 cm, outperforming CBT, ESINI, LRMA, N-only, and R-only by 40.02%, 44.70%, 40.00%, 41.97%, and 35.39%, respectively. As the packet loss rate increases from 10% to 30%, the MAE of

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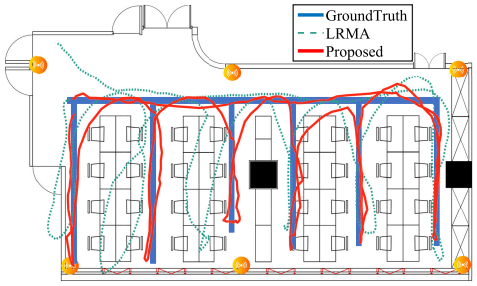


Fig. 7. Qualitative comparison of navigation performance between the proposed framework and the LRMA method in harsh environments.

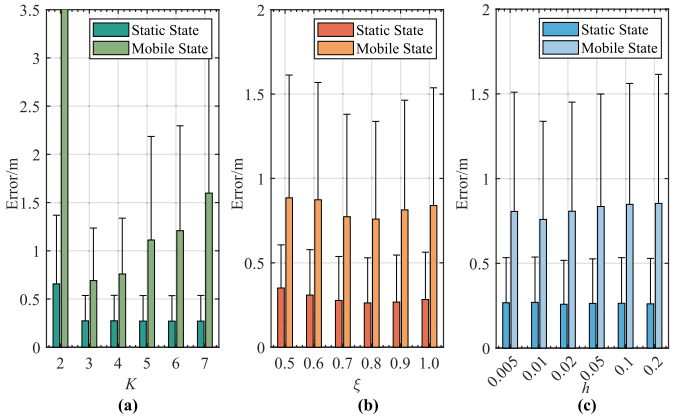


Fig. 8. Analysis of parameters. (a) Size of time window. (b) Threshold coefficient. (c) Smoothing parameter.

the proposed framework remains relatively stable, whereas the MAE80 exhibits a noticeable increase. Nevertheless, in Exp. 6, the proposed framework still maintains the best performance compared to SOTA methods, with MAE80 reductions of 53.74%, 31.71%, 45.22%, 37.18%, and 19.35% compared to CBT, ESINI, LRMA, N-only, and R-only.

Fig. 6(a) shows that under a low packet loss rate of approximately 10%, the proposed framework achieves the best cdf performance. Owing to the limited robustness of LRMA with rank 2 compared to GROUSE, the R-only model outperforms LRMA. At a higher loss rate of 30%, the proposed framework still performs best. Similar to the static case, the N-only model without packet loss recovery offers no clear advantage, whereas incorporating recovery makes the benefit of NLoS identification evident, yielding further gains. Fig. 6(c) compares the robustness of different packet loss rates in mobile scenarios. Under 10% packet loss, all methods achieve an average localization error within 1.2 m. At 30% packet loss, CBT and LRMA degrade significantly, while the proposed framework maintains consistently strong performance.

Fig. 7 compares the real-time navigation performance of LRMA and the proposed framework under 10% packet loss. As the target approaches the pillar, LRMA deviates from the ground-truth trajectory due to NLoS errors. In contrast, the proposed framework, which addresses both packet loss recovery and NLoS identification, maintains a more accurate navigation path. Additionally, it better handles lag during sudden directional changes, such as quick turns, compared to LRMA.

D. Parameter Performance Analysis

To illustrate the optimality and stability of the parameters in the proposed framework, we analyzed the time window size K, threshold coefficient ξ, and smoothing parameter h in both static and dynamic environments, keeping other parameters at their default values. The parameter performance analysis is conducted under a 30% packet loss rate using the Exp. 2 and Exp. 6 datasets.

Fig. 8(a) shows the impact of varying the time window size K on localization and navigation performance, with K ranging from 2 to 7. When $K = 2$, the estimated ToF matrix $\hat{\mathbf{A}}_k \in \mathbb{R}^{2\times 6}$. Since the preset rank equals 2, the factorization becomes full rank, which amplifies noise and degrades performance. Navigation performance peaks when K is set to 3 or 4, but declines with larger K due to the low ranging frequency (1 Hz) relative to the moving speed. When the window exceeds 5 frames (5 s), signal correlation weakens and linearity deteriorates, leading to trajectory lag or oversmoothing. In static scenarios, performance remains stable. For robustness, K is typically set to 4, though it should be adapted to the ranging frequency and target speed.

Fig. 8(b) illustrates the impact of the threshold coefficient ξ in the range of 0.5–1.0, showing similar trends in both static and mobile scenarios. When ξ is too small, NLoS identification is excessive, leading to the misclassification of LoS data. As ξ approaches 1, the framework behaves like the R-only model, resulting in performance degradation.

Fig. 8(c) presents the effects of the smoothing parameter h in the range of 0.005–0.2. Performance changes are not significant in the static dataset, but a clear trend is observed in the mobile dataset, with optimal performance achieved at $h = 0.01$. When h is too small, residuals are overclassified, weakening NLoS identification. Conversely, when h is too large, the classification ability of KDE is reduced, degrading performance to that of the R-only model.

V. Conclusion

This article proposed a continuous environment sensing ToF-based localization and navigation framework for harsh environments, which enables packet loss recovery and NLoS identification. The designed continuous environment sensing module integrates historical NLoS data and the target's motion states to dynamically update environmental characteristics, resulting in improved packet loss recovery and NLoS identification. We implemented the localization framework on commodity UWB devices and evaluated the system in harsh environments. Comprehensive experimental results and analysis demonstrate that under a 30% packet loss rate, the proposed framework attains average localization and navigation accuracies of 17.71 and 47.42 cm, respectively, and reduces localization error by 51.03% in static scenarios and 50.18% in mobile scenarios compared with SOTA methods. In future work, we aim to address the limitations of fallback-based strategies under prolonged sparse observations by formulating localization as a spatiotemporal sequence learning problem. Specifically, we plan to construct temporally

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aware input sequences of masked measurements and integrate attention mechanisms to capture long-range dependencies. This would enable the development of a self-supervised foundation model for robust end-to-end localization and navigation, particularly suited for extreme and dynamically changing environments.

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